

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-GA-1201A-03 Sheet 1 of 1
OPL Title	Pump-Motor Alignment Check (Laser)	Discipline	Mechanical	
Equipment	GA-1201A - HEXANE FEED PUMP	Area / Unit	1200 - HEXANE FEED SYSTEM	
Related Interlock	SEQ-1201 (SIL 1)	P&ID; Ref	TJC-LLD-PID-1201	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **pump-motor alignment check (laser)** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Shaft misalignment is the #1 root cause of GA-1201A high-vibration trips (VSHH-1201) and premature bearing/seal failure. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and LOTO the motor, remove coupling guard	Isolation & LOTO verified
2	Mount laser alignment brackets across the coupling spacer	Parts match BOM / spec
3	Rotate 0-90-180-270 deg and capture readings	Torque / clearance to spec
4	Target angular < 0.05 mm/100 mm and offset < 0.05 mm	No leak / smooth operation
5	Add/remove shims under motor feet, re-check, and record the alignment report	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr.	API Plan 11 flush orifice RO-1201 partially plugged causing seal face.	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v.
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to.
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Shaft misalignment is the #1 root cause of GA-1201A high-vibration trips (VSHH-1201) and premature bearing/seal failure.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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